Algorithm 1: $((\mathfrak{R}_g, \mathcal{D}_g, \mathcal{L}_g))_{g\in\mathcal{G}} \leftarrow \texttt{MultiGPUFamilyRollout}(\pi, \mathsf{m}, N_{\mathsf{seq}})$

/* The rollout profile is frozen and identical on all GPUs; queues and rollout state are GPU-local. */

Parameters: finite PBG $(\mathcal{S}, \mathcal{T}, \mathcal{P}, o, (\mathcal{E}_\sigma)_\sigma, \mathcal{X}, \mu, \chi)$; frozen $\pi = \pi^{\theta_k}$; $\mathsf{m} \in \{\mathsf{normal}, \mathsf{RB}, \mathsf{branched\text{-}RB}\}$; GPU set $\mathcal{G}$; N_{seq} sequential trajectory starts, unused in branched-RB

Parameters: finite execution-family priority poset $(\mathcal{M}, \leq_{\mathcal{M}})$; public routing map $\kappa : \{\sigma : o(\sigma) \in \mathcal{P}\} \to \mathcal{M}$; one curried forward implementation $\texttt{Forward}_m$ per m

Parameters: for branched-RB, a frozen public black-box intensity $\beta : \mathcal{S} \setminus \mathcal{T}^{\mathsf{pre-c}} \to [1, \infty)$

Hyperparameters: per-GPU hard memory limits; family batch-row limits; calibrated actual-cost physical forward planner; chunked arena and active-trajectory limits

Definitions: $\mathsf{seq} \leftarrow \mathbb{1}_{\mathsf{m}\neq\mathsf{branched\text{-}RB}}$; $\mathcal{T}_{\mathsf{m}} \leftarrow \mathcal{T}$ if $\mathsf{m} = \mathsf{normal}$, else $\mathcal{T}^{\mathsf{pre-c}}$; $\mathrm{Stop}_{\mathsf{m}}(\sigma) \leftarrow \mathbb{1}_{\sigma\in\mathcal{T}_{\mathsf{m}}}$

Definitions: $\mathrm{Round}(z; K) = \lfloor z \rfloor + \mathrm{Bernoulli}(z - \lfloor z \rfloor; K)$; $\iota(\mathsf{s})$ is the schedule- and placement-independent logical address of s: (global trajectory index, path) in sequential modes and (rank tree index, abstract tree path) in branched-RB. All draws use disjoint keys $K(k, \iota(\mathsf{s}), u, \mathsf{purpose})$

Definitions:
$$\mathbb{P}^{\mathfrak{r}_{\mathsf{s}}}(x \mid \sigma) = \frac{\mu(x)\prod_{q\in\mathcal{P}\cup\{\mathsf{c}\}} r_{\mathsf{s},q}(\sigma \mid x_q)}{\sum_{y\in\mathcal{X}} \mu(y)\prod_{q\in\mathcal{P}\cup\{\mathsf{c}\}} r_{\mathsf{s},q}(\sigma \mid y_q)}$$

Definitions: $\texttt{Pack}_{g,m}(\mathcal{Q})$ returns a stable insertion-order block containing only family m; it never subdivides or routes by street or any other public-stage label. $\mathrm{PhysicalPacks}_{g,m}(\mathcal{B})$ may sort the block's exact current-model rows by predicted padded work and split it into one or more safe physical forwards, with a device-resident inverse scatter back to the logical block. It may not change queue membership, logical order, sampling addresses, or the mixed-street contract. Feasibility counts one model row per distinct exact current-model input, the maximum padded forward cost within each physical pack, child sampling, and expansion workspace; $\phi_{g,m}(\mathcal{B}) \in [0, 1]$ is its calibrated fill; $\mathrm{load}_{g,m}(\mathcal{Q})$ is waiting work in full-batch units and $\mathrm{age}(\mathcal{Q})$ is the oldest wait time; $\mathrm{Full}_{g,m}(\mathcal{Q})$ holds when the packed batch reaches the family's full-batch threshold

Definitions: $\mathfrak{A}_g$ is a preallocated chunked struct-of-arrays arena addressed only by integer occurrence handles; it owns topology, state references, reach rows, RNG addresses, child spans, and the monotone status $\mathsf{NEW} \to \mathsf{QUEUED} \to \mathsf{EVALUATING} \to \mathsf{DONE}$. The family work queues, counters, pack metadata, inverse scatters, and reusable input/output/sampler staging slabs are device-resident. Stable-v1 permits a declared fixed-size device-to-host control block for each logical family decision (including an actual-cost vector when used), never one scalar/control round trip per physical pack; it is recorded as control-sync count, elapsed time, and bytes. Dynamic compaction, scalar tensor reads, and every undeclared scheduler synchronization remain forbidden

Definitions: $\mathrm{Enqueue}_g(\mathcal{P}')$ appends each $\mathsf{s} \in \mathcal{P}'$ to the bounded work queue $\mathcal{Q}_{g,\kappa(\ell_g(\mathsf{s}))}$ and atomically changes its status from NEW to QUEUED; $\mathrm{SafeAdmit}_g(a)$ reserves retained storage and the largest in-flight transaction before admitting a further sequential starts; $\max \varnothing \leftarrow 0$

Output: $\mathfrak{R}_g$: one labeled tree rooted at $\varnothing$ if branched-RB, or GPU-local sequential trajectory records otherwise; decision rows $\mathcal{D}_g$; stopped leaves $\mathcal{L}_g$

/* Poker specialization: $\mathcal{M} = \{\mathsf{pre}\} \sqcup \{\mathsf{post}_A : 2 \leq A \leq |\mathcal{P}|\}$, $\mathsf{pre} \prec \mathsf{post}_A$ for every A, and the post_A are pairwise incomparable. Street is ordinary row metadata: mixed streets in the same post_A call are padded and masked, never scheduled separately. */

Before sampling, deterministically partition $\{1, \ldots, N_{\mathsf{seq}}\} = \bigsqcup_{g\in\mathcal{G}} \mathcal{I}_g$ if $\mathsf{seq} = 1$, using calibrated per-GPU safe capacities and throughput (equal shares on homogeneous GPUs)

foreach $g \in \mathcal{G}$ ***in parallel*** **do**
- $\mathcal{D}_g, \mathcal{L}_g \leftarrow \varnothing$
- $(\mathcal{Q}_{g,m})_{m\in\mathcal{M}} \leftarrow \varnothing$
- **if** $\mathsf{seq} = 0$ **then**
  - $\mathfrak{S}_g \leftarrow \{\varnothing\}$
  - $\ell_g(\varnothing) \leftarrow \varnothing$
  - $\mathfrak{r}_{g,\varnothing} \leftarrow (1)_{q\in\mathcal{P}\cup\{\mathsf{c}\}}$
  - $\iota(\varnothing) \leftarrow (g, \varnothing)$
  - $\mathfrak{R}_g \leftarrow (\mathfrak{S}_g, \ell_g, N_g, \mathfrak{r}_g)$
  - $(\mathcal{P}_0, \mathcal{L}_0) \leftarrow \texttt{ChanceClose}_g(\{\varnothing\}, \mathsf{m})$
  - $\mathrm{Enqueue}_g(\mathcal{P}_0)$
  - $\mathcal{L}_g \leftarrow \mathcal{L}_g \sqcup \mathcal{L}_0$
  - $\mathcal{U}_g \leftarrow \varnothing$ /* exactly one root on this GPU */
- **else**
  - $\mathfrak{R}_g \leftarrow \varnothing$
  - $\mathcal{U}_g \leftarrow \mathcal{I}_g$ /* unstarted trajectory indices */
- **end**
- **while** $\mathcal{U}_g \neq \varnothing$ or $\bigsqcup_{m\in\mathcal{M}} \mathcal{Q}_{g,m} \neq \varnothing$ **do**
  - $\mathcal{F}_g \leftarrow \{m \in \mathcal{M} : \mathrm{Full}_{g,m}(\mathcal{Q}_{g,m})\}$
  - **if** $\mathcal{F}_g \neq \varnothing$ **then**
    - $m \leftarrow \arg\max_{m\in\min_{\leq_{\mathcal{M}}}\mathcal{F}_g} (\mathrm{load}_{g,m}(\mathcal{Q}_{g,m}), \mathrm{age}(\mathcal{Q}_{g,m}))$ /* lexicographic; upstream full work first */
  - **else if** $\mathsf{seq} = 1$ *and* $a_g \leftarrow \max\{a \leq |\mathcal{U}_g| : \mathrm{SafeAdmit}_g(a)\} > 0$ **then**
    - Remove a_g indices i from $\mathcal{U}_g$ in stable order and, in one device batch, allocate trajectory-start occurrences s_i^0 with $\ell_g(\mathsf{s}_i^0) = \varnothing$, $\mathfrak{r}_{g,\mathsf{s}_i^0} = (1)_q$, $X_i \sim \mu$ using $K(k, (i, \varnothing), 0, \mathsf{private})$, and $\iota(\mathsf{s}_i^0) = (i, \varnothing)$
    - Append these occurrences to $\mathfrak{R}_g$
    - $(\mathcal{P}_0, \mathcal{L}_0) \leftarrow \texttt{ChanceClose}_g(\{\mathsf{s}_i^0\}_i, \mathsf{m})$
    - $\mathrm{Enqueue}_g(\mathcal{P}_0)$
    - $\mathcal{L}_g \leftarrow \mathcal{L}_g \sqcup \mathcal{L}_0$
    - **continue**
  - **else**
    - assert($\bigsqcup_m \mathcal{Q}_{g,m} \neq \varnothing$)
    - $m \leftarrow \arg\max_{m:\mathcal{Q}_{g,m}\neq\varnothing} (\phi_{g,m}(\texttt{Pack}_{g,m}(\mathcal{Q}_{g,m})), \mathrm{age}(\mathcal{Q}_{g,m}), -h(m))$ /* forced partial drain; h is any topological height */
  - **end**
  - $\mathcal{B} \leftarrow \texttt{Pack}_{g,m}(\mathcal{Q}_{g,m})$, remove its handles from $\mathcal{Q}_{g,m}$, and atomically mark them $\mathsf{EVALUATING}$
  - $(\mathcal{J}_1, \ldots, \mathcal{J}_r) \leftarrow \mathrm{PhysicalPacks}_{g,m}(\mathcal{B})$
  - **foreach** $\mathcal{J}_j$ ***in one warmed family-m GPU call*** **do**
    - $\sigma_{\mathsf{s}} \leftarrow \ell_g(\mathsf{s})$ for $\mathsf{s} \in \mathcal{J}_j$
    - $p_{\mathsf{s}} \leftarrow o(\sigma_{\mathsf{s}})$ for $\mathsf{s} \in \mathcal{J}_j$
    - $\Pi_{\mathsf{s}} \leftarrow \texttt{Forward}_m(\sigma_{\mathsf{s}}, \mathfrak{r}_{g,\mathsf{s}}; \theta_k) = \left(\pi(\cdot \mid x_{p_{\mathsf{s}}}, \sigma_{\mathsf{s}}), V^{\theta_k}(x_{p_{\mathsf{s}}}, \sigma_{\mathsf{s}}), \ldots\right)_{x_{p_{\mathsf{s}}}\in\mathcal{X}_{p_{\mathsf{s}}}}$
  - **end**
  - $(\mathcal{A}, \mathcal{F}) \leftarrow \texttt{SampleChildren}_g(\mathcal{B}, (\Pi_{\mathsf{s}})_{\mathsf{s}\in\mathcal{B}}, \mathsf{m})$
  - $(\mathcal{P}', \mathcal{L}') \leftarrow \texttt{ChanceClose}_g(\mathcal{F}, \mathsf{m})$
  - $\mathcal{D}_g \leftarrow \mathcal{D}_g \sqcup \mathcal{A}$
  - $\mathrm{Enqueue}_g(\mathcal{P}')$
  - $\mathcal{L}_g \leftarrow \mathcal{L}_g \sqcup \mathcal{L}'$
  - Mark every occurrence in $\mathcal{B}$ DONE; only deterministic frozen outputs may be shared—every child keeps its own profile, event, reach, descendants, and later completion draws
  - assert($\bigsqcup_m \mathcal{Q}_{g,m} = \{$all live, non-in-flight player occurrences on GPU $g\}$)
  - assert(no chance occurrence is pending)
  - assert(hard structural and memory budgets hold)
- **end**
- **if** $\mathsf{seq} = 0$ **then**
  - assert($\mathfrak{S}_g$ is one tree with the unique root $\varnothing$)
- **end**

end

Impose no terminal barrier and perform no queue or model collective inside this routine: each GPU may proceed directly from its completed local rollout to local completion, backward target construction, and compaction; the first required collective is the normalization reduction in $\texttt{DistributedFixedRolloutUpdate}$

return $((\mathfrak{R}_g, \mathcal{D}_g, \mathcal{L}_g))_{g\in\mathcal{G}}$

Algorithm 2: $(\mathcal{A}, \mathcal{F}') \leftarrow \texttt{SampleChildren}_g(\mathcal{B}, \Pi, \mathsf{m})$

/* $\mathcal{B}$ is a player batch with curried output Π, or a chance batch with $\Pi = \bot$. */

Input: sampled occurrences $\mathcal{B} = (\mathsf{s}_j)_{j=1}^b$ of one owner kind; mode m; GPU-local rollout state and schedule-independent keys

Output: decision rows $\mathcal{A}$ (empty for chance); all sampled children $\mathcal{F}'$

$\mathcal{A}, \mathcal{F}' \leftarrow \varnothing$

for $j = 1, \ldots, b$ ***in one device batch*** **do**
- $\sigma_j \leftarrow \ell_g(\mathsf{s}_j)$
- $q_j \leftarrow o(\sigma_j)$
- $n_j \leftarrow \begin{cases} 1, & \mathsf{m} \neq \mathsf{branched\text{-}RB}, \\ \mathrm{Round}(\beta(\sigma_j); K(k, \iota(\mathsf{s}_j), 0, \mathsf{round})), & \mathsf{m} = \mathsf{branched\text{-}RB}; \end{cases}$
- **if** $\mathsf{m} = \mathsf{branched\text{-}RB}$ **then**
  - $N_g(\mathsf{s}_j) \leftarrow n_j$
- **end**

end

$\mathcal{I} \leftarrow \{(j, u) : 1 \leq j \leq b, 1 \leq u \leq n_j\}$ by prefix sum

reserve the exact child/edge block and maximum-feasible ragged sampler workspace

fail loudly rather than changing $(n_j)_j$ if a hard budget is exceeded

foreach $(j, u) \in \mathcal{I}$ ***in maximum-feasible device batches*** **do**
- $X_{j,u} \leftarrow \begin{cases} X_{i(j)}, & \mathsf{m} \neq \mathsf{branched\text{-}RB}, \\ X \sim \mathbb{P}^{\mathfrak{r}_{g,\mathsf{s}_j}}(\cdot \mid \sigma_j) \text{ with } K(k, \iota(\mathsf{s}_j), u, \mathsf{posterior}), & \mathsf{m} = \mathsf{branched\text{-}RB}; \end{cases}$
  /* the sibling evaluator intentionally returns a full joint profile */
- $E_{j,u} \sim \begin{cases} \pi(\cdot \mid X_{j,u,q_j}, \sigma_j) \text{ from } \Pi_j, & q_j \in \mathcal{P}, \\ \chi(\cdot \mid X_{j,u}, \sigma_j), & q_j = \mathsf{c}, \end{cases}$ using $K(k, \iota(\mathsf{s}_j), u, \mathsf{action})$ or $K(k, \iota(\mathsf{s}_j), u, \mathsf{chance})$, respectively
- allocate a distinct abstract child $\mathsf{s}_j\mathsf{a}_{j,u}$ and set $\iota(\mathsf{s}_j\mathsf{a}_{j,u}) = (\iota(\mathsf{s}_j), u)$, $\ell_g(\mathsf{s}_j\mathsf{a}_{j,u}) = \sigma_j E_{j,u}$
- $\mathfrak{r}_{g,\mathsf{s}_j\mathsf{a}_{j,u},v} \leftarrow \mathfrak{r}_{g,\mathsf{s}_j,v}$ for $v \neq q_j$
- $\mathfrak{r}_{g,\mathsf{s}_j\mathsf{a}_{j,u},q_j} \leftarrow \mathrm{Normalize}\left(\mathfrak{r}_{g,\mathsf{s}_j,q_j} \odot \begin{cases} \pi(E_{j,u} \mid -, \sigma_j), & q_j \in \mathcal{P}, \\ \chi(E_{j,u} \mid -, \sigma_j), & q_j = \mathsf{c} \end{cases}\right)$
- append $\mathsf{s}_j\mathsf{a}_{j,u}$ to $\mathfrak{S}_g$ in branched-RB, or to its trajectory record otherwise, and append it to $\mathcal{F}'$

end

if $\Pi \neq \bot$ **then**
- $\mathcal{A} \leftarrow \{(\mathsf{s}_j, (\mathsf{a}_{j,u}, E_{j,u})_{u=1}^{n_j}, n_j, \mathfrak{r}_{g,\mathsf{s}_j}, \mathrm{Old}(\Pi_j, (E_{j,u})_{u=1}^{n_j})) : 1 \leq j \leq b\}$ /* retain only loss-required old policy/value slices */

end

Discard branched-RB posterior profiles $X_{j,u}$

return $(\mathcal{A}, \mathcal{F}')$

Algorithm 3: $(\mathcal{P}', \mathcal{L}') \leftarrow \texttt{ChanceClose}_g(\mathcal{F}, \mathsf{m})$

/* Close every chance-only segment before returning control to the model-family scheduler. */

Input: new sampled occurrences $\mathcal{F}$; mode m; GPU-local rollout state

Output: next player occurrences $\mathcal{P}'$ and stopped leaves $\mathcal{L}'$

$\mathcal{P}', \mathcal{L}' \leftarrow \varnothing$

while $\mathcal{F} \neq \varnothing$ **do**
- Partition $\mathcal{F}$ into $\mathcal{L}_0 = \{\mathsf{s} : \mathrm{Stop}_{\mathsf{m}}(\ell_g(\mathsf{s})) = 1\}$, $\mathcal{P}_0 = \{\mathsf{s} : \mathrm{Stop}_{\mathsf{m}}(\ell_g(\mathsf{s})) = 0, o(\ell_g(\mathsf{s})) \in \mathcal{P}\}$, and $\mathcal{C}_0 = \{\mathsf{s} : \mathrm{Stop}_{\mathsf{m}}(\ell_g(\mathsf{s})) = 0, o(\ell_g(\mathsf{s})) = \mathsf{c}\}$
- assert($|\mathcal{F}| = |\mathcal{L}_0| + |\mathcal{P}_0| + |\mathcal{C}_0|$)
- $\mathcal{P}' \leftarrow \mathcal{P}' \sqcup \mathcal{P}_0$
- $\mathcal{L}' \leftarrow \mathcal{L}' \sqcup \mathcal{L}_0$
- **if** $\mathsf{m} = \mathsf{branched\text{-}RB}$ **then**
  - $N_g(\mathsf{s}) \leftarrow 0$ for $\mathsf{s} \in \mathcal{L}_0$
- **end**
- Partition $\mathcal{C}_0$ into stable maximum-feasible chance-sampling batches $(\mathcal{C}_v)_v$
- $\mathcal{F} \leftarrow \varnothing$
- **foreach** $\mathcal{C}_v$ **do**
  - $(\varnothing, \mathcal{F}_v) \leftarrow \texttt{SampleChildren}_g(\mathcal{C}_v, \bot, \mathsf{m})$
  - $\mathcal{F} \leftarrow \mathcal{F} \sqcup \mathcal{F}_v$
- **end**

end

return $(\mathcal{P}', \mathcal{L}')$